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COS326 – Database Systems

Pyspark Practical Exercise

This practical introduces you to Apache Spark programming using PySpark within Google Colab. You will perform data ingestion, transformation, and analysis on a dataset of student exam scores. By completing this exercise, you will gain hands-on experience with Spark DataFrames, grouping, aggregation, and joins.

1. Setup

a) Open Google Colab and create a new Python notebook.

b) Copy and paste the following code block to install PySpark and initialize the Spark environment:

```
!pip install pyspark
!pip install findspark

import findspark
findspark.init()

from pyspark.sql import SparkSession
spark = SparkSession.builder.appName("Student Score
Analysis").getOrCreate()
```

2. Create Dataset

Run the following code block to create a CSV file of student exam scores:

```
import pandas as pd

data = {
    "Name": [
        "Thabo", "Naledi", "Sipho", "Ayanda",
"Kabelo",
        "Lerato", "Siyanda", "Zanele", "Marlize",
"Nomsa",
        "Thabo", "Naledi", "Sipho", "Ayanda",
"Kabelo",
        "Lerato", "Siyanda", "Zanele", "Marlize",
"Nomsa",
        "Thabo", "Naledi", "Sipho", "Ayanda",
"Kabelo",
        "Lerato", "Siyanda", "Zanele", "Marlize",
"Nomsa"
    ],
    "Subject": [
        "Maths", "Maths", "Maths", "Maths",
"Maths",
        "Maths", "Maths", "Maths", "Maths",
"Maths",
        "English", "English", "English", "English",
"English",
        "English", "English", "English", "English",
"English",
        "Science", "Science", "Science",
        "Science", "Science", "Science",
        "Science", "Science", "Science", "Science"
    ],
    "Score": [
        88, 72, 94, 81, 77, 85, 90, 73, 79, 84,
        91, 76, 88, 95, 83, 80, 87, 78, 85, 92,
        96, 82, 89, 93, 85, 90, 97, 88, 91, 84
    ]
}

df_pd = pd.DataFrame(data)
df_pd.to_csv("students.csv", index=False)
```

3. Load and Display Data

Load the CSV into a Spark DataFrame and preview the data:

```
df = spark.read.csv("students.csv", header=True,
inferSchema=True)
df.show(10)
```

4. Analysis Tasks

Perform the following analyses on the dataset using PySpark functions:

a) Compute the average score per subject.

b) Find the maximum score per subject.

c) Join the results to identify the top scorer for each subject.

```
from pyspark.sql.functions import avg, max, col

avg_scores =
df.groupBy("Subject").agg(avg("Score").alias("Avera
geScore"))
max_scores =
df.groupBy("Subject").agg(max("Score").alias("MaxSc
ore"))

df_alias = df.alias("d")
max_alias = max_scores.alias("m")

top_scorers = (
    df_alias.join(
        max_alias,
        (col("d.Subject") == col("m.Subject")) &
        (col("d.Score") == col("m.MaxScore"))
    )
    .select(col("d.Name").alias("Name"),
col("d.Subject").alias("Subject"),
col("d.Score").alias("Score"))
)

avg_scores.show()
max_scores.show()
top_scorers.show()
```

5. Homework

1. Add a new subject (e.g., 'Physics') and assign random scores for each student.
2. Compute the median score per subject using PySpark functions (read more here).
3. Create a bar chart (using Matplotlib or Seaborn) to visualize average scores per subject.
4. Export the results (average and top scorers) to new CSV files.